

Sumin Jung

📍 Seoul, Republic of Korea ✉️ sumin00716@gmail.com 📞 +82-10-6487-8026 in suminjung716 🌐 jasmine00716

Research Interests

Large-scale multimodal and foundation models for medical AI; vision-language learning; medical image analysis; clinical representation learning across institutions, modalities, and tasks.

Education

Korea University, Seoul, Republic of Korea

Mar 2023 – Feb 2025

M.S. in Computer Engineering

- Advisor: Prof. Jin Tae Kwak; GPA: 4.0/4.5
- Selected coursework: Medical Imaging; Efficient AI and Hardware Design; Advanced Digital Signal Processing; Random Signal Analysis

Dongduk Women's University, Seoul, Republic of Korea

Mar 2019 – Feb 2023

B.S. in Computer Science

- GPA: 3.72/4.5

Research Experience

PathRadX: Pathology Foundation Models for Radiology

2025

Contributing Researcher, DeepClue Inc.

- Supported data preparation, result analysis, and manuscript drafting for a study evaluating whether a frozen pathology foundation model could transfer to radiology tasks.
- Contributed to experiments spanning MURA, RSNA pneumonia, and MedFMC datasets and comparisons of modality-adaptation and task-classification strategies; collaborators led model design, implementation, and experiment execution.
- Published in the MedAGI 2025 workshop proceedings held with MICCAI 2025.

ISLES 2024: Multimodal Infarct Segmentation

2024

Research Design and Analysis, Korea University

- Contributed problem definition, data preparation, method and experimental design, comparative analysis, and abstract preparation for a 3D Vision Transformer using NCCT, CTP, CTA, and Tmax inputs.
- Collaborated with a teammate who implemented the code and ran the experiments.

3D-MoReT: Collateral Imaging from 4D MR Perfusion

2023 – 2024

First Author and Lead Researcher, Korea University

[Code](#) 

- Led the study from problem formulation and data preprocessing through model design, implementation, experiments, comparative analysis, and initial manuscript drafting.
- Designed a 3D CNN-Transformer regression model combining MobileNetV2, MobileViT, ViT, and residual/skip connections to generate five-phase collateral maps from 4D DSC-MRP.
- Evaluated the model on 952 patients from two university hospitals; outperformed similarly sized models and matched substantially heavier baselines.
- Published the work as first author in the *International Journal of Computer Assisted Radiology and Surgery* and presented it at CARS 2024.

Industry Experience

DeepClue Inc.

Sep 2024 – Jun 2026

Team Leader, Engineering & Manufacturing Team

- Led the translation of collateral-imaging research into DeepClue-Precise MRI/CT software as a medical device (SaMD), coordinating clinical requirements, system architecture, implementation, verification, regulatory documentation, and team operations.
- Developed the MRI application and image-processing pipeline end to end—including DICOM preprocessing, image analysis, GUI, output generation, and testing
- Designed the reusable application framework adopted by the CT product and collaborated with the developer responsible for its CT-specific core image-processing pipeline.
- Designed and executed software verification and validation, prepared regulatory and quality-system documents, coordinated with hospitals and external organizations, and led activities supporting MFDS Class I medical-device

registration.

- Supported hospital prototype deployment and converted clinical feedback into software requirements and validation criteria.

Publications

-
- | | |
|--|---------------------|
| H. J. Ki, H. G. Roh, S. B. Lee, J. T. Kwak, J. J. Park, Y. S. Jeon, S. Jung , H. Yang, C. Yu, J. S. Lee, H. J. Lee, and H. J. Kim. “Collateral Perfusion Score for Predicting Futile Endovascular Thrombectomy in Acute Anterior Circulation Stroke.” <i>Korean Journal of Radiology</i> . <i>Under review</i> . | <i>Under Review</i> |
| H. Kim, H. G. Roh, J. T. Kwak, H. J. Ki, I. S. Kim, S. Jung , H. Yang, J. J. Park, Y. S. Jeon, and H. J. Kim. “Simultaneous generation of color-coded arteriography, venography, and dynamic angiography: methodology and clinical applications in stroke.” <i>La Radiologia Medica</i> , 130(11):1820–1826. doi:10.1007/s11547-025-02074-z ↗ . | 2025 |
| H. Yang, S. Jung , and J. T. Kwak. “From pathology to radiology: evaluating the applicability of pathology foundation models.” In <i>Foundation Models for General Medical AI (MedAGI 2025)</i> , Lecture Notes in Computer Science, vol. 16112, pp. 34–43, Springer. doi:10.1007/978-3-032-07845-2_4 ↗ . | 2025 |
| H. J. Ki, H. G. Roh, J. T. Kwak, I. S. Kim, J. J. Park, Y. S. Jeon, H. Yang, S. Jung , J. S. Lee, and H. J. Kim. “The CT collateral map: collateral perfusion estimation and baseline lesion assessment after acute anterior circulation ischemic stroke.” <i>La Radiologia Medica</i> , 130(2):235–247. doi:10.1007/s11547-024-01941-5 ↗ . | 2025 |
| Y. J. Jeon, H. G. Roh, S. Jung , H. Yang, H. J. Ki, J. J. Park, T.-J. Lee, N. I. Shin, J. S. Lee, J. T. Kwak, and H. J. Kim. “Clinical feasibility of deep learning-driven magnetic resonance angiography collateral map in acute anterior circulation ischemic stroke.” <i>Scientific Reports</i> , 15, Article 2304. doi:10.1038/s41598-025-85731-7 ↗ . | 2025 |
| E. de la Rosa, R. Su, M. Reyes, et al. (including H. Yang, J. T. Kwak, and S. Jung). “ISLES’24: Final Infarct Prediction with Multimodal Imaging and Clinical Data. Where Do We Stand?” <i>arXiv:2408.10966</i> . arxiv.org/abs/2408.10966 ↗ . <i>Preprint</i> . | 2024 (rev. 2025) |
| S. Jung , H. Yang, H. J. Kim, H. G. Roh, and J. T. Kwak. “3D mobile regression vision transformer for collateral imaging in acute ischemic stroke.” <i>International Journal of Computer Assisted Radiology and Surgery</i> , 19(10):2043–2054. doi:10.1007/s11548-024-03229-5 ↗ . | 2024 |
| H. Yang, S. Jung , V. T. L. Trinh, B. C. Doanh, J. Wang, H. G. Roh, H. J. Kim, and J. T. Kwak. “Investigation of Deep Learning Models and Training Strategies for Brain Stroke Segmentation.” <i>IEIE Annual Conference</i> . | 2024 |

Honors & Awards

-
- **Best Poster Award**, 38th Annual Conference of the Korean Society of Medical Imaging Informatics, 2024
 - **9th Place**, Ischemic Stroke Lesion Segmentation Challenge (ISLES 2024), 2024
 - **Honorable Mention**, Samsung AI Challenge, Image Quality Assessment, 2023
 - **Gold Prize**, Joint Capstone Design Competition, 2022

Patent

-
- | | |
|---|-----------------|
| Hyun Yang, Sumin Jung , and Hong Gee Roh. “Machine Learning Method for Generating Hypoperfusion Lesion Segmentation Maps from Cerebral Collateral Blood Flow Images Obtained Using Multi-Phase Dynamic Images, and a Method and System for Predicting Ischemic Penumbra Area Using Generated Hypoperfusion Lesion Segmentation Maps.” <i>Republic of Korea Patent Application No. 10-2024-0117934</i> . Co-inventor . | <i>Aug 2024</i> |
| ◦ Co-led development of the 3D medical-image framework for generating hypoperfusion lesion segmentation maps using anatomical information. | |

Teaching Experience

-
- | | |
|---|--------------------|
| Machine Learning and Intelligence for Electrical and Electronic Engineers
<i>Teaching Assistant, Korea University</i> | <i>Spring 2023</i> |
| ◦ Ran lab sessions. | |
| ◦ Answered student questions and provided academic support. | |